

- check what "slide" formats are used for TB ! Initial Multi-well
- check what conditions TB require $\rightarrow$ Nominal (want to expand)
 - $\rightarrow$ During body reaction
- Check what Medium is used for growth (depends on cells - want flow)
- Check what each 3 commercial solutions provide (Become experts)
- explore other products
- do some literature review for any project

want possibility for CO_2 control

- ultimately $\rightarrow$ want fully contained solution (Ideally)

To do: go through options and indicate how they work a bit more.

- Go back through, and find out operating conditions (e.g. Vibrations, etc) Basically can it work in a BSL3 fume hood
- try find sourcing for tubes, inline filters for air / fluids (perfusion)

- Temp changes physical properties
- O^2 is required for live cell activities (duh)
- CO^2 is required for pH control! (didn't know)

↳ [according to culturecollections.org.uk → Incubators use 5% CO^2]

for pH of 7.4 at physiological temp in human body
a 20:1 ratio of bicarbonate to carbonic acid must
be maintained (confirm for non body)

[* from www.culturecollections.org.uk]

Physiological pH → 7.2 → 7.4 (7.4 generally?)
(confirm!)

EMEM with EBSS
needs 5% CO_2 Incubator
→ for pH 7.0-7.4

* EMEM with EBSS (medium) use 26mM of sodium bicarbonate

* MEM uses 44mM of sodium bicarbonate

→ 25% → 11% CO_2

$CO^2 \uparrow \Rightarrow pH \downarrow$

$CO^2 \downarrow \Rightarrow pH \uparrow$

New to do.

- Put together a brief onboarding document for project 2
- Organise researched info
- 1st step - fully contained chamber
- Put together a short (like 2 slide) power point
- Mainly work with plate

- is autoclave in future head?

1. Ask for reqs
2. Come up with concept designs + specs
3. Present them
4. Get feedback
5. Research
6. Repeat!

more specifically

- Find out what Mandy / Robyn / Justin wants (product, concept, Research, etc)
- Juggle the act of Research, documentation, concept design as in the numbered list above
- Communicate with people to better get a sense of what their idea of the project is

- For now: 1. Finish this onboarding documents
- Finalize Specs! for Project → 2. Ask them to see if its good, is it what they want
3. Confirm that plan is to use multi-well plates
ask what lighting they want / need (does top need to be clear?)
4. Come up with some ideas (many)
5. Present them

96 well plate:

Length: 127.8 ± 0.4 mm

Width: 89.5 ± 0.4 mm

height w lid: 16.6 ± 0.2 mm

wo lid: 14.4 ± 0.2 mm